

ECO 362: Financial Investments

Bonds

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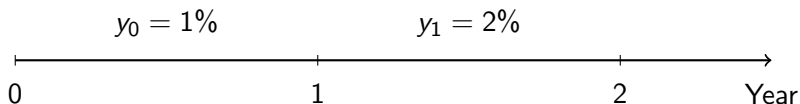
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Understanding bond markets

- What determines bond yields?
- In particular, what determines the slope of the term structure of interest rates?
- Various types of bonds.
 - Treasury securities (bills, notes, and bonds).
 - Treasury Inflation Protected Securities (TIPS).
 - Corporate bonds.
- **Question:** If the 5-year yield is 1.95% and the 10-year yield is 2.94%, which is a better investment?

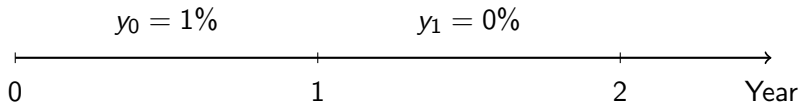
Term structure when there is no uncertainty

- 1-year continuously compounded yield is
 - 1% in year 0.
 - 2% in year 1.



- What is the 2-year yield in year 0?
 - $\frac{1}{2}(1\% + 2\%) = 1.5\%$
- Thus, the term structure is upward sloping in year 0.

- 1-year continuously compounded yield is
 - 1% in year 0.
 - 0% in year 1.



- What is the 2-year yield in year 0?
 - $\frac{1}{2}(1\% + 0\%) = 0.5\%$
- Thus, the term structure is downward sloping in year 0.

Federal Reserve

- Federal Open Market Committee (FOMC) consists of 7 members of the Board of Governors and 5 Reserve Bank presidents.
- Meets 8 times a year to set the fed funds target rate.
- Fed funds rate is the overnight riskless interest rate between banks.
- Open market operations (purchases/sales of Treasury securities) ensure that the fed funds rate is near the target rate.

FOMC statement on September 17, 2025

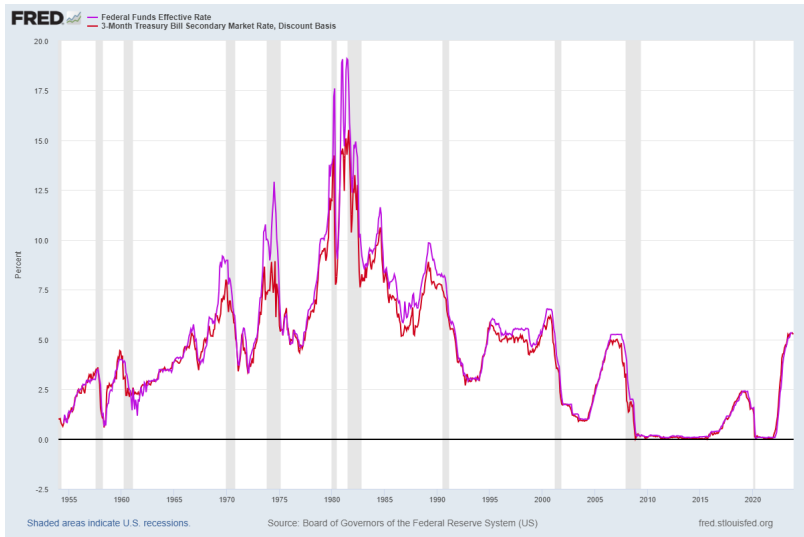
In support of its goals and in light of the shift in the balance of risks, the Committee decided to lower the target range for the federal funds rate by 1/4 percentage point to 4 to 4-1/4 percent... The Committee is strongly committed to supporting maximum employment and returning inflation to its 2 percent objective...

The Committee would be prepared to adjust the stance of monetary policy as appropriate if risks emerge that could impede the attainment of the Committee's goals. The Committee's assessments will take into account a wide range of information, including readings on labor market conditions, inflation pressures and inflation expectations, and financial and international developments.

Parsing the FOMC statement

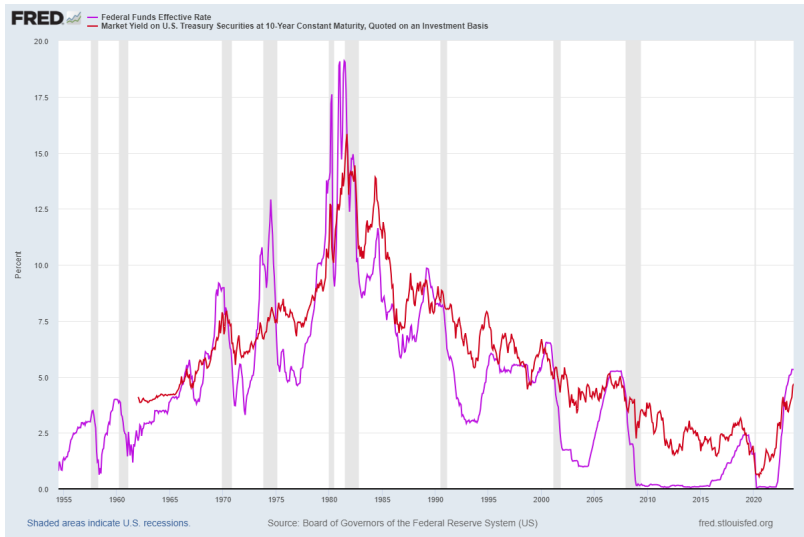
- Sets the current fed funds rate: “the Committee decided to lower the target range for the federal funds rate by 1/4 percentage point to 4 to 4-1/4 percent”
- Forward guidance: “The Committee would be prepared to adjust the stance of monetary policy as appropriate if risks emerge that could impede the attainment of the Committee’s goals.”
- Long-term bond yields reflect expectation of future interest rates.

Fed funds rate and 3-month T-bill



Source: FRED Economic Data (9/30/2023)

Fed funds rate and 10-year Treasury



Source: FRED Economic Data (9/30/2023)

Three key facts

1. Mean reversion: Fed funds rate has a tendency to rise when low and fall when high.
 2. Level factor: Riskless interest rates of all maturities move up/down together, anchored by the fed funds rate.
 3. Short-term rates (e.g., 3 months) are more volatile than long-term rates (e.g., 10 years).
- A term structure model explains all 3 facts.

Zero-coupon bonds

- **Continuous compounding** throughout this lecture.
- Index time (years) as $s = 0, 1, \dots$
- Index maturity (years) as $t = 1, \dots, T$.
- Term structure of interest rates (or zero-coupon yield curve) today: $y_0(1), \dots, y_0(T)$.
- Recall the relation between price and yield.

$$P_0(t) = \exp(-ty_0(t))$$

- How are the yields at different maturities related to each other?

Definition of return

- Buy a t -year bond today at price $P_0(t)$.
- Sell it a year from now at price $P_1(t-1)$.
- Return from year 0 to 1 is

$$\exp(r_1(t)) = \frac{P_1(t-1)}{P_0(t)} = \frac{\exp(-(t-1)y_1(t-1))}{\exp(-ty_0(t))}$$

- In logarithms,

$$r_1(t) = ty_0(t) - (t-1)y_1(t-1)$$

- Rearrange as

$$ty_0(t) = y_0(1) + r_1(t) - y_0(1) + (t-1)y_1(t-1)$$

Expectations hypothesis

- Take expectation of the previous equation.

$$ty_0(t) = y_0(1) + \mathbb{E}_0[r_1(t) - y_0(1)] + \mathbb{E}_0[(t-1)y_1(t-1)]$$

- Expectations hypothesis: Assume that expected excess returns are constant in maturity and time.

$$\mathbb{E}_0[r_1(t) - y_0(1)] = \mu$$

- Under the expectations hypothesis,

$$ty_0(t) = y_0(1) + \mu + \mathbb{E}_0[(t-1)y_1(t-1)]$$

Term structure under the expectations hypothesis

- For a 2-year bond,

$$\begin{aligned}2y_0(2) &= y_0(1) + \mu + \mathbb{E}_0[y_1(1)] \\ y_0(2) &= \frac{1}{2}\mathbb{E}_0[y_0(1) + y_1(1)] + \frac{1}{2}\mu\end{aligned}$$

- For a 3-year bond,

$$\begin{aligned}3y_0(3) &= y_0(1) + \mu + \mathbb{E}_0[2y_1(2)] \\ &= y_0(1) + \mu + \mathbb{E}_0[y_1(1) + \mu + \mathbb{E}_1[y_2(1)]] \\ y_0(3) &= \frac{1}{3}\mathbb{E}_0[y_0(1) + y_1(1) + y_2(1)] + \frac{2}{3}\mu\end{aligned}$$

- The general expression is

$$y_0(t) = \underbrace{\frac{1}{t} \sum_{s=0}^{t-1} \mathbb{E}_0[y_s(1)]}_{\text{expected short rate}} + \underbrace{\frac{t-1}{t} \mu}_{\text{term premium}}$$

- Expected short rate: t -year bonds reflect the expected path of the short rate over the next t years.
- Why forward guidance is powerful.
- Term premium is constant under the expectations hypothesis.

A simple model of the short rate

- Assume that the short rate follows a mean-reverting process.

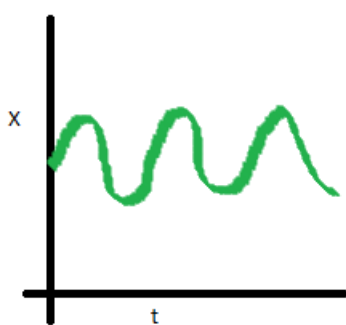
$$y_s(1) - \bar{y} = \rho(y_{s-1}(1) - \bar{y}) + \epsilon_s$$

where $\rho \leq 1$.

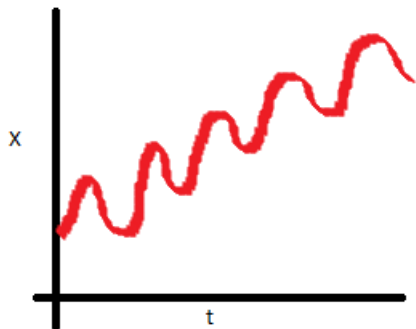
- $\bar{y}$ is the long-run mean of the short rate.
 - ρ determines the speed of mean reversion.
 - ϵ_s are mean-zero shocks that are uncorrelated over time.
- When $\rho = 1$, the short rate follows a random walk.

$$y_s(1) = y_{s-1}(1) + \epsilon_s$$

Mean-reverting process vs. random walk



Stationary series



Non-Stationary series

Expectation of the short rate

- Expectation 1 year ahead.

$$\begin{aligned}\mathbb{E}_0[y_1(1) - \bar{y}] &= \mathbb{E}_0[\rho(y_0(1) - \bar{y}) + \epsilon_1] \\ &= \rho(y_0(1) - \bar{y})\end{aligned}$$

- Expectation 2 years ahead.

$$\begin{aligned}\mathbb{E}_0[y_2(1) - \bar{y}] &= \mathbb{E}_0[\rho^2(y_0(1) - \bar{y}) + \rho\epsilon_1 + \epsilon_2] \\ &= \rho^2(y_0(1) - \bar{y})\end{aligned}$$

- How about infinitely far in the future (if $\rho < 1$)?

$$\mathbb{E}_0[y_\infty(1) - \bar{y}] = \rho^\infty(y_0(1) - \bar{y}) = 0$$

- In the long-run, the short rate is expected to revert to the mean $\bar{y}$ (Fact 1).

Term structure model

- The t -year yield is

$$y_0(t) - \bar{y} = \frac{1}{t} \sum_{s=0}^{t-1} \mathbb{E}_0[y_s(1) - \bar{y}] + \frac{t-1}{t} \mu$$

- If the short-rate follows a mean-reverting process,

$$\begin{aligned} \sum_{s=0}^{t-1} \mathbb{E}_0[y_s(1) - \bar{y}] &= (1 + \rho + \rho^2 + \cdots + \rho^{t-1})(y_0(1) - \bar{y}) \\ &= \left(\frac{1 - \rho^t}{1 - \rho} \right) (y_0(1) - \bar{y}) \end{aligned}$$

- Thus, the t -year yield depends only on the 1-year yield.

$$y_0(t) - \bar{y} = \underbrace{\frac{1}{t} \left(\frac{1 - \rho^t}{1 - \rho} \right)}_{b(t)} (y_0(1) - \bar{y}) + \underbrace{\frac{t-1}{t} \mu}_{a(t)}$$

Key implications of the term structure model

- Explains the level factor (**Fact 2**).
 - The t -year yield moves together with the 1-year yield. In fact, perfectly correlated.
- Explains why short-term yields are more volatile than long-term yields (**Fact 3**).
 - Standard deviation of the t -year yield is

$$\sigma(y_0(t)) = \underbrace{b(t)}_{<1} \sigma(y_0(1))$$

- Because of mean reversion, there is less uncertainty about the long-run expected short rate (must equal $\bar{y}$).

Slope of the term structure

- Yield spread between t -year and 1-year bonds is

$$y_0(t) - y_0(1) = \underbrace{(b(t) - 1)}_{<0} (y_0(1) - \bar{y}) + \frac{t-1}{t} \mu$$

- Slope depends on 2 effects.
 - Expectation of future short rates.
 - If $y_0(1) > \bar{y}$, the short rate is expected to fall. Makes the term structure more downward sloping.
 - If $y_0(1) < \bar{y}$, the short rate is expected to rise. Makes the term structure more upward sloping.
 - Term premium makes the term structure more upward sloping.

Yield is not the expected return

- **Question:** If the 5-year yield is 1.95% and the 10-year yield is 2.94%, which is a better investment?
- Under the expectations hypothesis, they have the same expected return!
- An upward-sloping term structure means that the short rate is expected to rise.
- A higher yield is offset by a lower expected capital gain as interest rates rise.

$$\begin{aligned}\mathbb{E}_0[r_1(t)] &= y_0(1) + \mu \\ &= \underbrace{y_0(t)}_{\text{yield}} + \underbrace{(t-1)\mathbb{E}_0[y_0(t) - y_1(t-1)]}_{\text{capital gain}}\end{aligned}$$

Example

- Suppose that $y_0(1) = 0.90\%$ and $\mu = 0.33\%$.
- 5-year bond with yield 1.95%.

$$\begin{aligned}\mathbb{E}_0[r_1(5)] &= 0.90\% + 0.33\% \\ &= \underbrace{1.95\%}_{\text{yield}} - \underbrace{0.72\%}_{\text{capital loss}}\end{aligned}$$

- 10-year bond with yield 2.94%.

$$\begin{aligned}\mathbb{E}_0[r_1(10)] &= 0.90\% + 0.33\% \\ &= \underbrace{2.94\%}_{\text{yield}} - \underbrace{1.71\%}_{\text{capital loss}}\end{aligned}$$

Inflation

- Treasury securities are nominal bonds that promise to pay \$1 in the future.
- Purchasing power of \$1 in the future is uncertain.
- Inflation is defined in terms of the consumer price index (CPI), which is the dollar cost of a basket of goods and services at a given point in time.
- Inflation from time 0 to 1 is

$$\exp(i_1) = \frac{CPI_1}{CPI_0}$$

Nominal vs. real return

- Let y_0 be the nominal yield at time 0.
- Real return r_1 from time 0 to 1 is unknown because of inflation uncertainty.

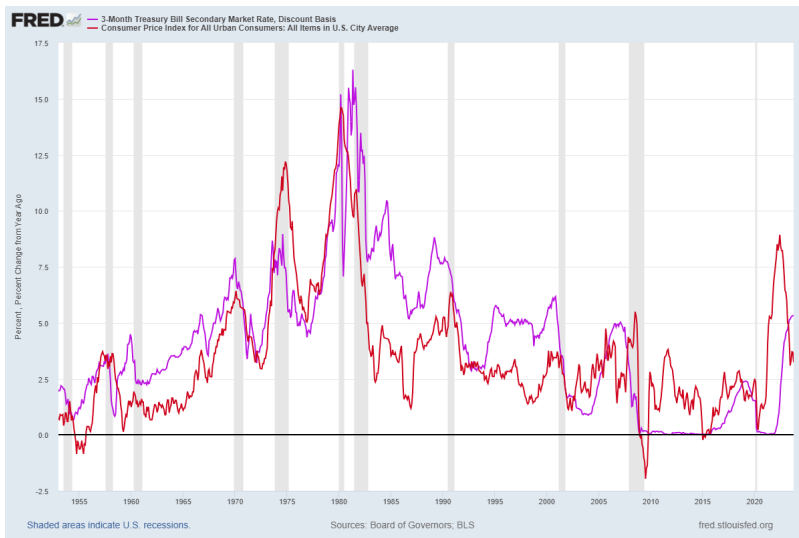
$$r_1 = y_0 - i_1$$

- Fisher equation:

$$y_0 = \underbrace{\mathbb{E}[r_1]}_{\text{expected real return}} + \underbrace{\mathbb{E}[i_1]}_{\text{expected inflation}}$$

- If expected real return is relatively constant over time, higher expected inflation implies a higher nominal yield.

3-month T-bills and inflation



Source: FRED Economic Data (9/30/2023)

Treasury Inflation Protected Securities

- US Treasury issues maturities of 5, 10, and 30 years.
- Face value is indexed to the CPI (with a 3 month lag).

$$F_t = \frac{\text{CPI}_t}{\text{CPI}_0} = \exp(ti_t)$$

- Pays semi-annual coupon of cF_t on each coupon date t .
- Because the coupons and principal are indexed to the CPI, there is no inflation risk (ignoring the 3-month lag).

Yield on TIPS

- T -year TIPS with coupon rate c and face value \$1 at issuance.
- Price is related to the yield through

$$P_{TIPS}(T) = \sum_{t=1}^T c \exp(-ty_{TIPS}(T)) + \exp(-Ty_{TIPS}(T))$$

- T : Maturity.
- $P_{TIPS}(T)$: Price.
- $y_{TIPS}(T)$: Yield to maturity.
- Yield on TIPS is interpreted as the real interest rate.

Zero-coupon TIPS

- Price of T -year zero-coupon TIPS.

$$\begin{aligned}
 P_{TIPS}(T) &= \exp(-Ty_{TIPS}(T)) \\
 &= \mathbb{E}[M_T F_T] = \mathbb{E}[M_T \exp(Ti_T)] \\
 &= \underbrace{\mathbb{E}[M_T]}_{\exp(-Ty_b(T))} \mathbb{E}[\exp(Ti_T)] + \text{Cov}(M_T, \exp(Ti_T))
 \end{aligned}$$

where $y_b(T)$ is the nominal zero-coupon yield.

- Divide through by $\mathbb{E}[M_T]$.

$$\frac{\exp(Ty_b(T))}{\exp(Ty_{TIPS}(T))} = \mathbb{E}[\exp(Ti_T)] \left(1 + \text{Cov} \left(\frac{M_T}{\mathbb{E}[M_T]}, \frac{\exp(Ti_T)}{\mathbb{E}[\exp(Ti_T)]} \right) \right)$$

Break-even inflation

- In logarithms,

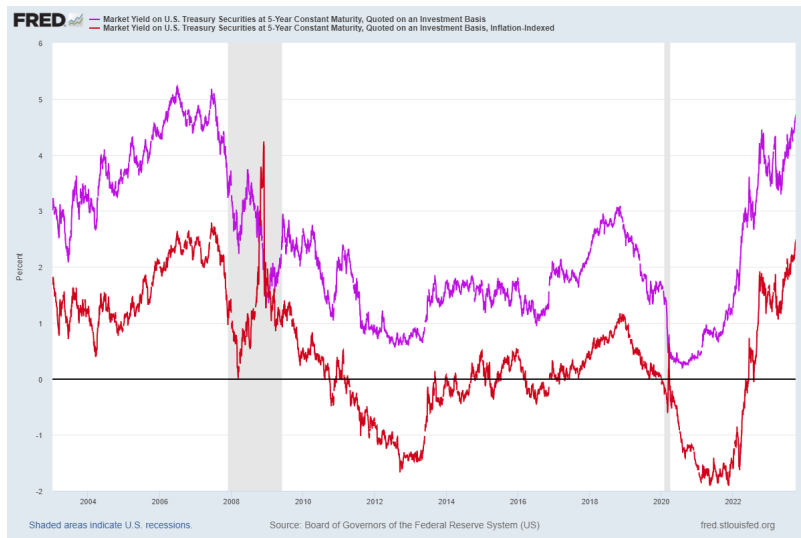
$$\underbrace{y_b(T) - y_{TIPS}(T)}_{\text{break-even inflation}} = \underbrace{\frac{1}{T} \ln(\mathbb{E}[\exp(Ti_T)])}_{\text{expected inflation}} + \text{IRP}$$

- Break-even inflation: Yield spread between nominal Treasury and TIPS.
- Inflation risk premium:

$$\text{IRP} = \frac{1}{T} \ln \left(1 + \text{Cov} \left(\frac{M_T}{\mathbb{E}[M_T]}, \frac{\exp(Ti_T)}{\mathbb{E}[\exp(Ti_T)]} \right) \right)$$

- Zero if inflation risk is idiosyncratic (i.e., uncorrelated with the SDF).
- If inflation is positively correlated with the SDF, nominal Treasuries are risky and require an extra risk premium.

5-year yield on TIPS and nominal Treasuries



Source: FRED Economic Data (9/30/2023)

Credit ratings

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Bond Ratings									
		Very High Quality		High Quality		Speculative		Very Poor	
Standard & Poor's		AAA	AA	A	BBB	BB	B	CCC	D
Moody's		Aaa	Aa	A	Baa	Ba	B	Caa	C
At times both Moody's and Standard & Poor's have used adjustments to these ratings: S&P uses plus and minus signs: A+ is the strongest A rating and A− the weakest. Moody's uses a 1, 2, or 3 designation, with 1 indicating the strongest.									
Moody's	S&P								
Aaa	AAA	Debt rated Aaa and AAA has the highest rating. Capacity to pay interest and principal is extremely strong.							
Aa	AA	Debt rated Aa and AA has a very strong capacity to pay interest and repay principal. Together with the highest rating, this group comprises the high-grade bond class.							
A	A	Debt rated A has a strong capacity to pay interest and repay principal, although it is somewhat more susceptible to the adverse effects of changes in circumstances and economic conditions than debt in higher-rated categories.							
Baa	BBB	Debt rated Baa and BBB is regarded as having an adequate capacity to pay interest and repay principal. Whereas it normally exhibits adequate protection parameters, adverse economic conditions or changing circumstances are more likely to lead to a weakened capacity to pay interest and repay principal for debt in this category than in higher-rated categories. These bonds are medium-grade obligations.							
Ba	BB	Debt rated in these categories is regarded, on balance, as predominantly speculative with respect to capacity to pay interest and repay principal in accordance with the terms of the obligation. BB and Ba indicate the lowest degree of speculation, and CC and Ca the highest degree of speculation. Although such debt will likely have some quality and protective characteristics, these are outweighed by large uncertainties or major risk exposures to adverse conditions. Some issues may be in default.							
B	B								
Caa	CCC								
Ca	CC								
C	C	This rating is reserved for income bonds on which no interest is being paid.							
D	D	Debt rated D is in default, and payment of interest and/or repayment of principal is in arrears.							

Source: Bodie, Kane, and Marcus (2018, Figure 14.8)

Yield on corporate bonds

- T -year corporate bond with coupon rate c and face value \$1.
- Price is related to the yield through

$$P_{CB}(T) = \sum_{t=1}^T c \exp(-ty_{CB}(T)) + \exp(-Ty_{CB}(T))$$

- T : Maturity.
- $P_{CB}(T)$: Price.
- $y_{CB}(T)$: Yield to maturity.
- Credit risk: A firm may not pay all coupons and principal if it defaults.
- Higher credit risk implies lower price and higher yield.
- Thus, corporate bonds with low credit ratings are sometimes referred to as high-yield bonds.

Zero-coupon corporate bond

- In the event of default, a share D_T of face value is repaid.
- Price of a T -year zero-coupon corporate bond.

$$\begin{aligned}P_{CB}(T) &= \exp(-Ty_{CB}(T)) \\&= \mathbb{E}[M_T D_T] \\&= \underbrace{\mathbb{E}[M_T]}_{\exp(-Ty_b(T))} \mathbb{E}[D_T] + \text{Cov}(M_T, D_T)\end{aligned}$$

where $y_b(T)$ is the riskless zero-coupon yield.

- Divide through by $\mathbb{E}[M_T]$.

$$\frac{\exp(Ty_b(T))}{\exp(Ty_{CB}(T))} = \mathbb{E}[D_T] \left(1 + \text{Cov} \left(\frac{M_T}{\mathbb{E}[M_T]}, \frac{D_T}{\mathbb{E}[D_T]} \right) \right)$$

Credit spread

- In logarithms,

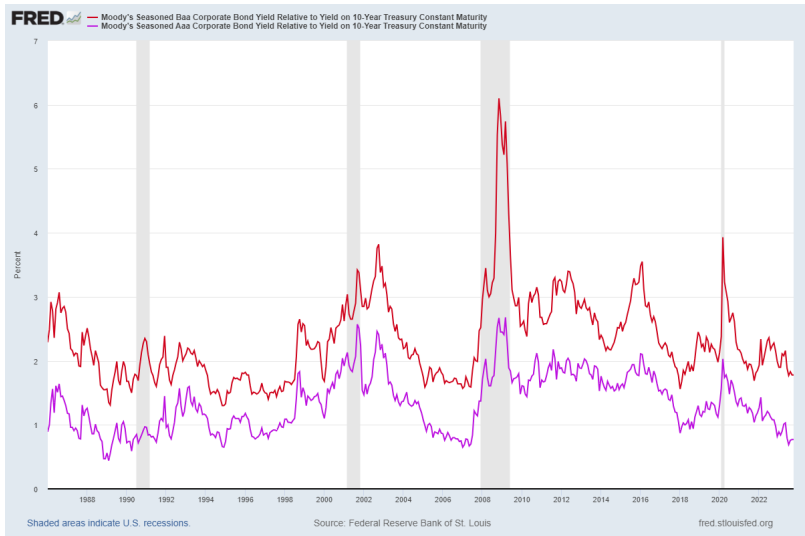
$$\underbrace{y_{CB}(T) - y_b(T)}_{\text{credit spread}} = \underbrace{-\frac{1}{T} \ln(\mathbb{E}[D_T])}_{\text{expected default}} + \text{CRP}$$

- Credit spread: Yield spread between corporate and Treasury bonds.
- Credit risk premium:

$$\text{CRP} = -\frac{1}{T} \ln \left(1 + \text{Cov} \left(\frac{M_T}{\mathbb{E}[M_T]}, \frac{D_T}{\mathbb{E}[D_T]} \right) \right)$$

- Zero if default risk is idiosyncratic (i.e., uncorrelated with the SDF).
- If default is positively correlated with the SDF, corporate bonds are risky and require an extra risk premium.

Corporate credit spreads



Source: FRED Economic Data (9/30/2023)

Credit spreads and expected excess returns

Corporate Bond Returns Jan 1987 to Dec 2011						
	AAA	Aa	A	Baa	Caa	Trsys
Yields						
Mean	6.20%	6.35%	6.72%	7.35%	15.45%	5.26%
Corp Spread	0.93%	1.09%	1.45%	2.09%	10.19%	
Excess Returns over Treasuries						
Mean	0.32%	0.42%	0.43%	1.04%	0.86%	
Stdev	2.61%	2.92%	4.00%	4.70%	16.00%	
Sharpe	0.124	0.143	0.109	0.221	0.054	

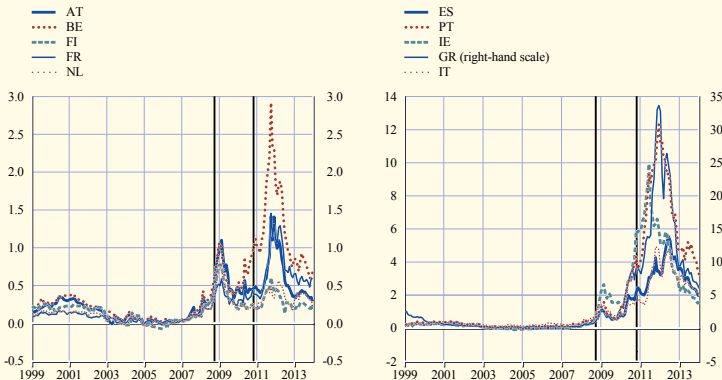
Source: Ang (2014, Table 9.13)

- Low-beta anomaly: Slightly higher expected returns on high-yield bonds, but not high enough relative to risk.

Sovereign credit spreads

Chart 1 Ten-year government bond yield spreads against Germany

(percentage points; January 1999 to February 2014)



Source: Bloomberg.

Notes: The first vertical bar denotes the bankruptcy of Lehman Brothers on 15 September 2008. The second vertical bar denotes the Deauville agreement on 18 October 2010.

Source: ECB Monthly Bulletin (May 2014)

Summary

- Slope of the term structure reflects expectation of future short rate.
- Break-even inflation: Expected inflation plus inflation risk premium.
- Credit spread: Expected default plus credit risk premium.